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1 C:\Users\sumer\Desktop\Projects\Python\test\venv\Scripts\python
  .exe C:\Users\sumer\Desktop\Projects\Python\test\Forest.py
2 Classification report for all hyperparameter combinations:
3 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 1, '
  min_samples_split': 2, 'n_estimators': 100}
4 Mean test score: 0.814 (std: 0.005)
5 Classification report:
6           precision    recall  f1-score   support
7
8      0           0.84      0.95      0.89       7040
9      1           0.67      0.35      0.46       1960
10
11     accuracy                0.82       9000
12    macro avg           0.75      0.65      0.68       9000
13 weighted avg           0.80      0.82      0.80       9000
14
15 =====
16 =====
17 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 1, '
  min_samples_split': 2, 'n_estimators': 200}
18 Mean test score: 0.815 (std: 0.005)
19 Classification report:
20           precision    recall  f1-score   support
21
22      0           0.84      0.95      0.89       7040
23      1           0.67      0.35      0.46       1960
24
25     accuracy                0.82       9000
26    macro avg           0.75      0.65      0.68       9000
27 weighted avg           0.80      0.82      0.80       9000
28
29 =====
30 =====
31 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 1, '
  min_samples_split': 5, 'n_estimators': 100}
32 Mean test score: 0.814 (std: 0.005)
33 Classification report:
34           precision    recall  f1-score   support
35
36      0           0.84      0.95      0.89       7040
37      1           0.67      0.35      0.46       1960
38
39     accuracy                0.82       9000
40    macro avg           0.75      0.65      0.68       9000
41 weighted avg           0.80      0.82      0.80       9000
42
43 =====
44 =====

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42 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 1, '
    min_samples_split': 5, 'n_estimators': 200}
43 Mean test score: 0.815 (std: 0.005)
44 Classification report:
45           precision    recall  f1-score   support
46
47      0           0.84       0.95       0.89       7040
48      1           0.67       0.35       0.46       1960
49
50   accuracy                0.82       9000
51   macro avg           0.75       0.65       0.68       9000
52   weighted avg           0.80       0.82       0.80       9000
53
54 =====
55 =====
56 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 2, '
    min_samples_split': 2, 'n_estimators': 100}
57 Mean test score: 0.814 (std: 0.005)
58 Classification report:
59           precision    recall  f1-score   support
60
61      0           0.84       0.95       0.89       7040
62      1           0.67       0.35       0.46       1960
63
64   accuracy                0.82       9000
65   macro avg           0.75       0.65       0.68       9000
66   weighted avg           0.80       0.82       0.80       9000
67
68 =====
69 =====
70 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 2, '
    min_samples_split': 2, 'n_estimators': 200}
71 Mean test score: 0.815 (std: 0.005)
72 Classification report:
73           precision    recall  f1-score   support
74
75      0           0.84       0.95       0.89       7040
76      1           0.67       0.35       0.46       1960
77
78   accuracy                0.82       9000
79   macro avg           0.75       0.65       0.68       9000
80   weighted avg           0.80       0.82       0.80       9000
81
82 =====
83 =====
84 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 2, '
    min_samples_split': 5, 'n_estimators': 100}
85 Mean test score: 0.815 (std: 0.005)
```

83 Classification report:

		precision	recall	f1-score	support
	0	0.84	0.95	0.89	7040
	1	0.67	0.35	0.46	1960
	accuracy			0.82	9000
	macro avg	0.75	0.65	0.68	9000
	weighted avg	0.80	0.82	0.80	9000

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94 Hyperparameters: {'max_depth': 5, 'min_samples_leaf': 2, 'min_samples_split': 5, 'n_estimators': 200}

95 Mean test score: 0.815 (std: 0.005)

96 Classification report:

		precision	recall	f1-score	support
	0	0.84	0.95	0.89	7040
	1	0.67	0.35	0.46	1960
	accuracy			0.82	9000
	macro avg	0.75	0.65	0.68	9000
	weighted avg	0.80	0.82	0.80	9000

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107 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}

108 Mean test score: 0.817 (std: 0.003)

109 Classification report:

		precision	recall	f1-score	support
	0	0.84	0.95	0.89	7040
	1	0.67	0.35	0.46	1960
	accuracy			0.82	9000
	macro avg	0.75	0.65	0.68	9000
	weighted avg	0.80	0.82	0.80	9000

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120 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200}

121 Mean test score: 0.818 (std: 0.003)

122 Classification report:

		precision	recall	f1-score	support
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125	0	0.84	0.95	0.89	7040
126	1	0.67	0.35	0.46	1960
127					
128	accuracy			0.82	9000
129	macro avg	0.75	0.65	0.68	9000
130	weighted avg	0.80	0.82	0.80	9000

131

132 =====
=====

133 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 1, '
min_samples_split': 5, 'n_estimators': 100}

134 Mean test score: 0.819 (std: 0.003)

135 Classification report:

136		precision	recall	f1-score	support
137					
138	0	0.84	0.95	0.89	7040
139	1	0.67	0.35	0.46	1960

140

141	accuracy			0.82	9000
142	macro avg	0.75	0.65	0.68	9000
143	weighted avg	0.80	0.82	0.80	9000

144

145 =====
=====

146 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 1, '
min_samples_split': 5, 'n_estimators': 200}

147 Mean test score: 0.819 (std: 0.003)

148 Classification report:

149		precision	recall	f1-score	support
150					
151	0	0.84	0.95	0.89	7040
152	1	0.67	0.35	0.46	1960

153

154	accuracy			0.82	9000
155	macro avg	0.75	0.65	0.68	9000
156	weighted avg	0.80	0.82	0.80	9000

157

158 =====
=====

159 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 2, '
min_samples_split': 2, 'n_estimators': 100}

160 Mean test score: 0.819 (std: 0.004)

161 Classification report:

162		precision	recall	f1-score	support
163					
164	0	0.84	0.95	0.89	7040
165	1	0.67	0.35	0.46	1960

166

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167     accuracy                0.82      9000
168   macro avg                0.75      0.65      0.68      9000
169   weighted avg              0.80      0.82      0.80      9000
170
171 =====
172   =====
173 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 2, '
    min_samples_split': 2, 'n_estimators': 200}
174 Mean test score: 0.819 (std: 0.004)
175 Classification report:
176
177           precision    recall  f1-score   support
178
179      0           0.84      0.95      0.89       7040
180      1           0.67      0.35      0.46       1960
181
182
183   accuracy                0.82      9000
184   macro avg                0.75      0.65      0.68      9000
185   weighted avg              0.80      0.82      0.80      9000
186
187 =====
188   =====
189 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 2, '
    min_samples_split': 5, 'n_estimators': 100}
190 Mean test score: 0.819 (std: 0.004)
191 Classification report:
192
193           precision    recall  f1-score   support
194
195      0           0.84      0.95      0.89       7040
196      1           0.67      0.35      0.46       1960
197
198
199   accuracy                0.82      9000
200   macro avg                0.75      0.65      0.68      9000
201   weighted avg              0.80      0.82      0.80      9000
202
203 =====
204   =====
205 Hyperparameters: {'max_depth': 10, 'min_samples_leaf': 2, '
    min_samples_split': 5, 'n_estimators': 200}
206 Mean test score: 0.819 (std: 0.004)
207 Classification report:
208
209           precision    recall  f1-score   support
210
211      0           0.84      0.95      0.89       7040
212      1           0.67      0.35      0.46       1960
213
214
215   accuracy                0.82      9000
216   macro avg                0.75      0.65      0.68      9000
217   weighted avg              0.80      0.82      0.80      9000

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210 =====
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211 Best hyperparameters found: {'max_depth': 10, '
min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators':
100}

212 Random Forest Classification Best Accuracy: 0.8211111111111111

213

214 Process finished with exit code 0

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